LabVIEW 엔지니어가 알아야 할 신기술 : RIO FPGA

한국내쇼날인스트루먼트
산업용 장비개발 솔루션 담당 윤 주 영 대리

이것으로 무엇을 할 수 있을까요?



뭐든 만들 수 있습니다.



LEGO World







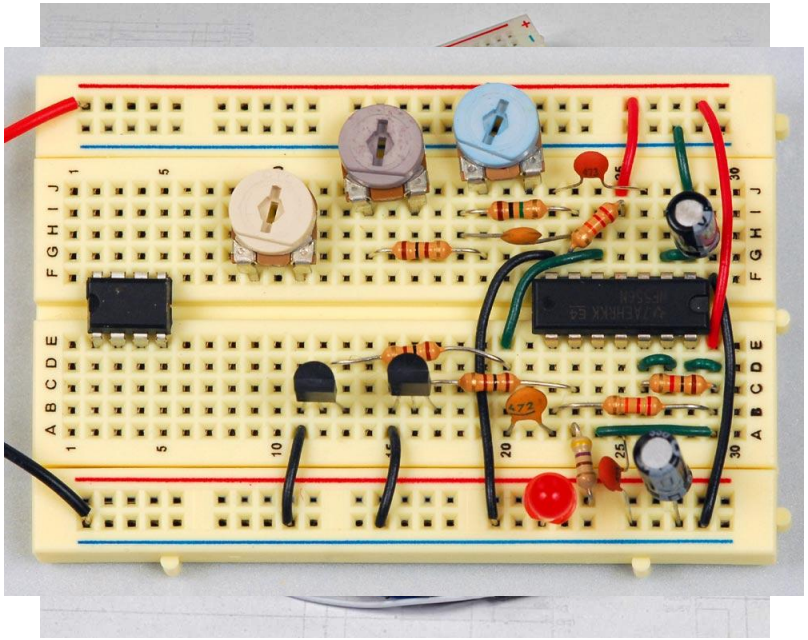

LEGO = FPGA ➔ RIO

FPGA (Field Programmable Gate Array)

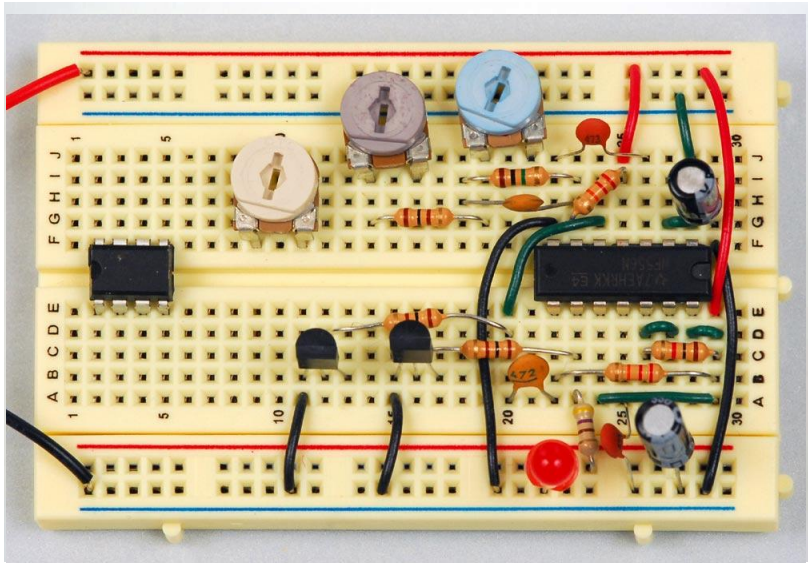
- 사용자가 언제든 원하는데로 프로그래밍과 재설정이 가능한 거대한 게이트 배열

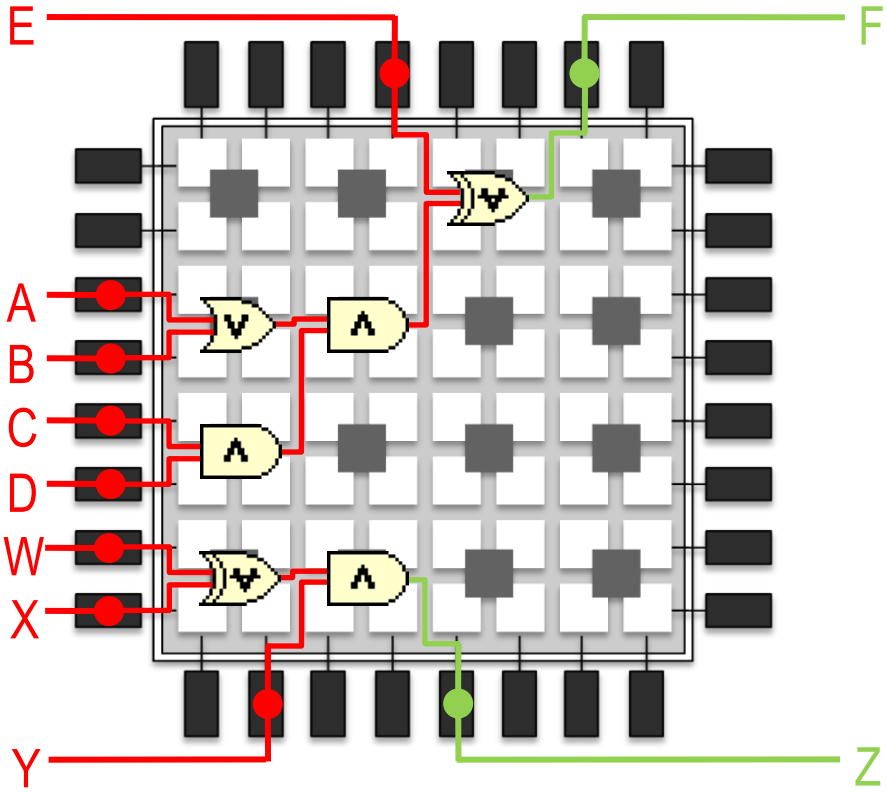






FPGA

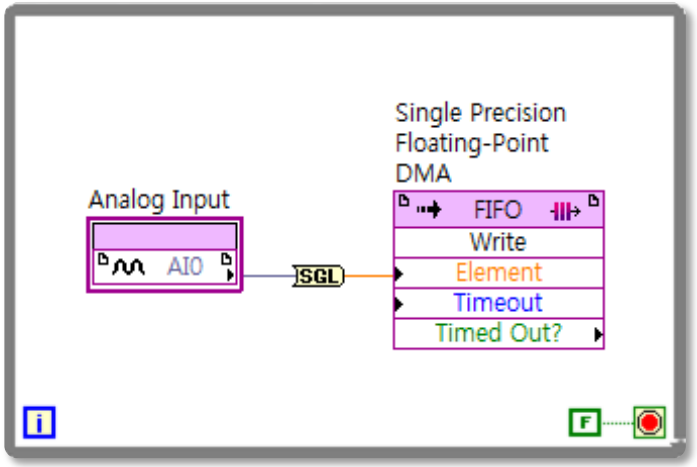




VHDL

LabVIEW FPGA

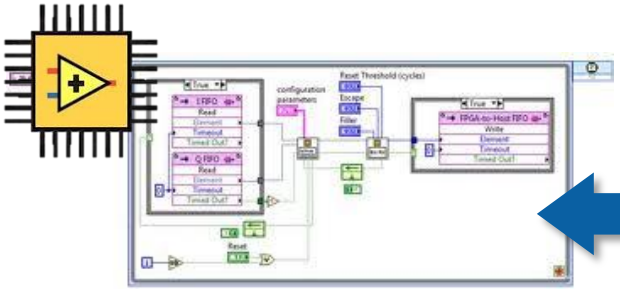
아날로그 입력 DMA 스트리밍



FPGA ➔ RIO

- Reconfigurable Input Output
  - FPGA를 상용 하드웨어에 적용
  - 사용자가 직접 FPGA에 접근 가능

  - LabVIEW를 이용하여 손쉽게 FPGA 사용







반제품 vs. 완제품



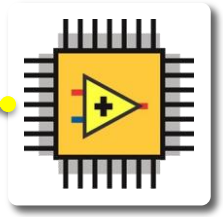






NI RIO의 역사

17년간 RIO 기술의 연구 & 개발



LabVIEW FPGA Module
첫 공개 & 데모시연



High-Performance CompactRIO
Single-Board RIO 2.0



Multi-Function
Single-Board RIO 2.0



FPGA 기반 임베디드 제어기
CompactRIO

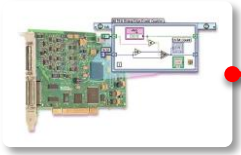


Single-Board RIO 1.0



SoC Zynq와 Linux기반의
임베디드 제어기 cRIO-9068

97 98 99 00 01 02 03 04 05 06 07 08 09 10 11 12 13



FPGA 기반 HW
PCI R 시리즈



IF-RIO



FlexRIO



세계 최초의 RF
Vector Signal Transceiver

LabVIEW RIO 하드웨어

CompactRIO and NI Single-Board RIO



Value



Ultra Rugged



Performance

PXI, PC RIO (R Series, NI FlexRIO)



High Performance

Expansion I/O



MXI-Express RIO



Ethernet RIO

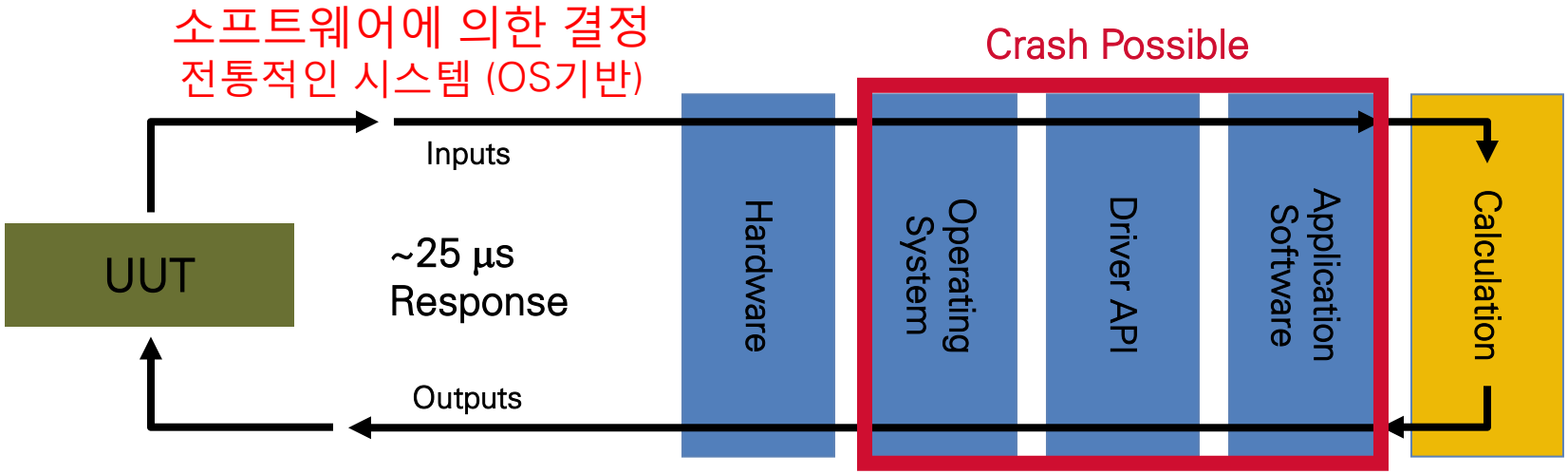


EtherCAT RIO

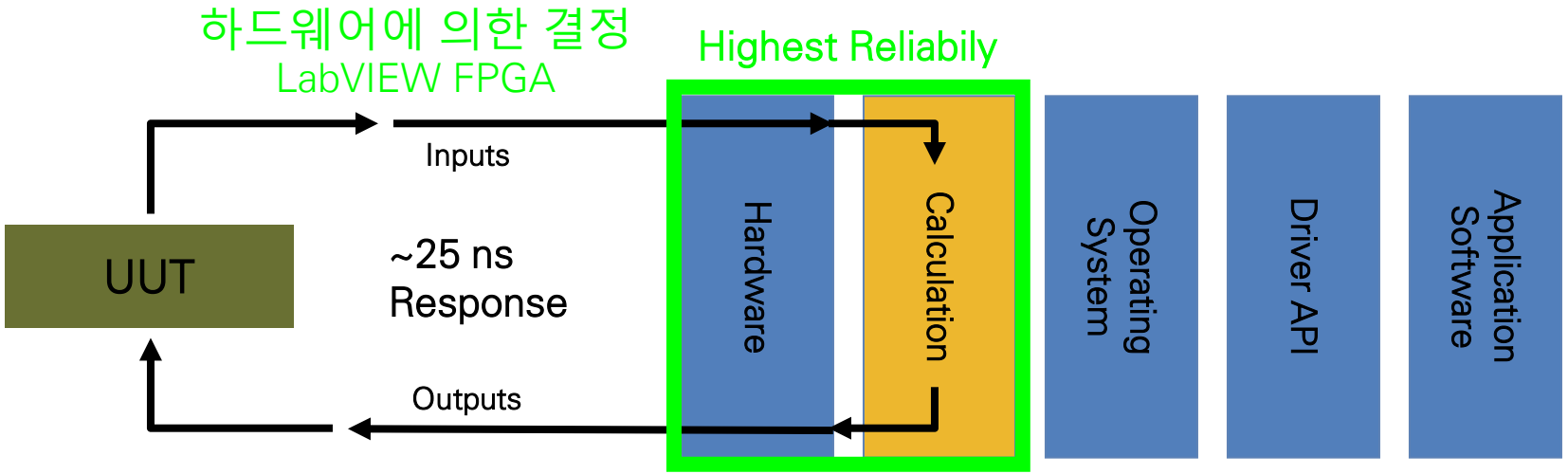


Wireless

RIO 기술의 특징 – 신뢰성

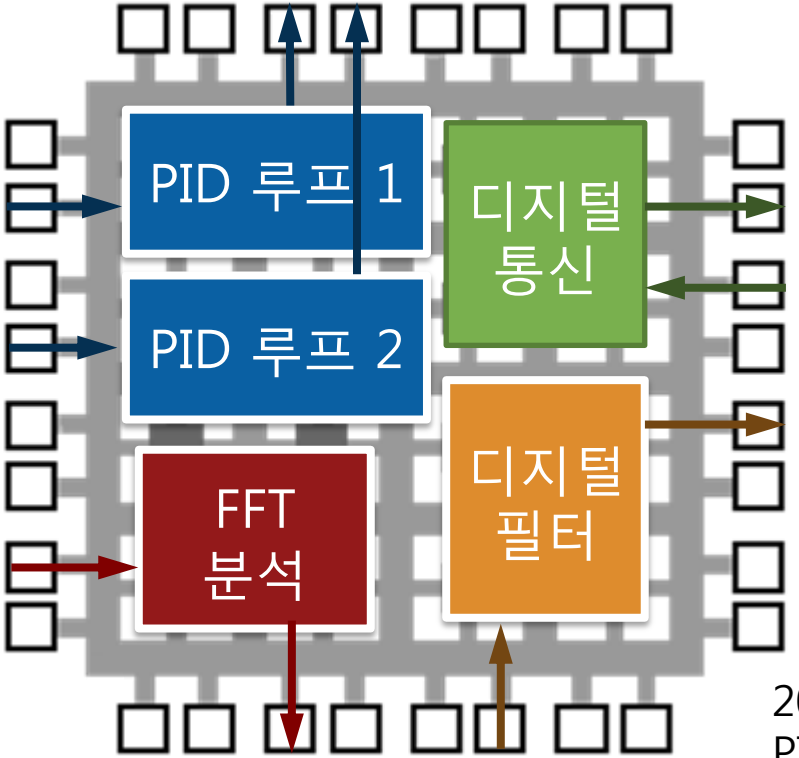


RIO 기술의 특징 – 신뢰성



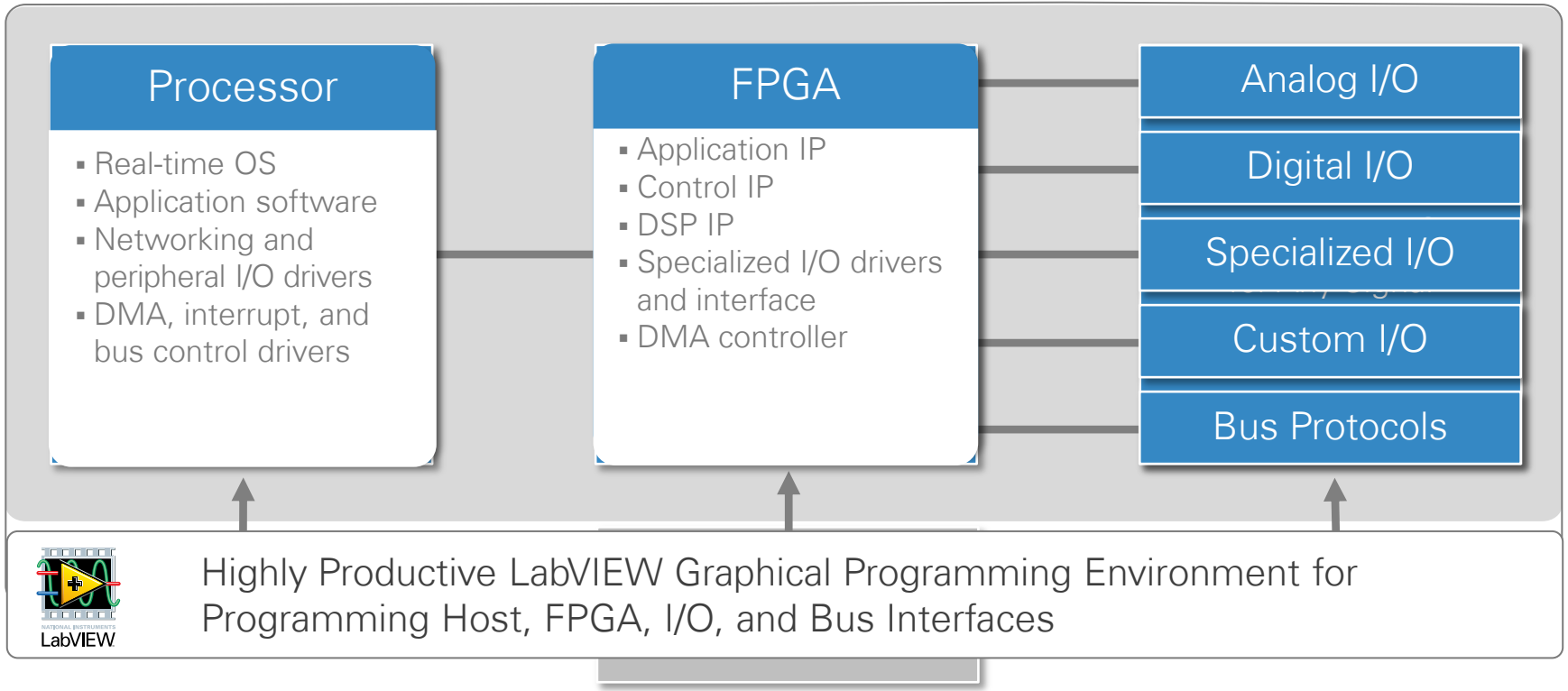
RIO 기술의 특징 – 병렬 프로세싱

- 사용자의 용도에 맞게 다양하게 재구성 가능
- ni.com/ipnet 유용한 예제 제공



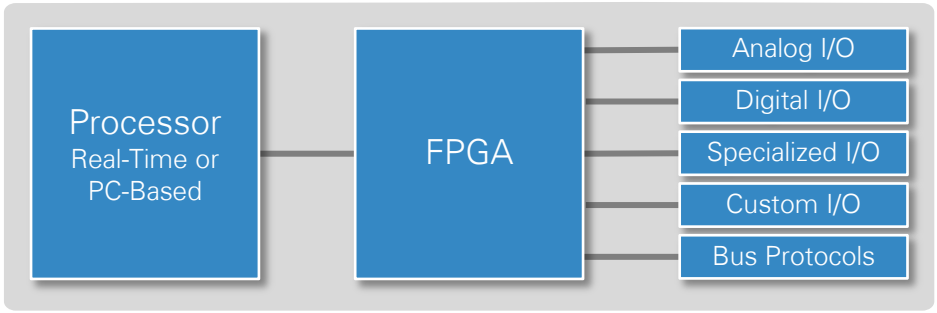
20개 이상의 FPGA 기반 PID 루프 동시 제어 가능

NI 하드웨어의 구성


Processor
▪ Real-time OS
▪ Application software
▪ Networking and peripheral I/O drivers
▪ DMA, interrupt, and bus control drivers
FPGA
▪ Application IP
▪ Control IP
▪ DSP IP
▪ Specialized I/O drivers and interface
▪ DMA controller
Analog I/O
Digital I/O
Specialized I/O
Custom I/O
Bus Protocols
LabVIEW
Highly Productive LabVIEW Graphical Programming Environment for Programming Host, FPGA, I/O, and Bus Interfaces

최고의 솔루션 제공

LabVIEW RIO Architecture.






Communication


Motion


Wireless


HMIs


Third-Party I/O


Vision

Alliance Partners

IP for Control, Analysis, etc.

NATIONAL INSTRUMENTS LabVIEW™

Q&A